SSH Keys for Cyfronet: One-Page Handout

Full guide: README | Example config: config.example

1. First login and known_hosts

- `ssh plguser@ares.cyfronet.pl`: check the fingerprint against the official docs, then `yes`.
- The host is saved in `~/.ssh/known_hosts`. Look it up: `ssh-keygen -F ares.cyfronet.pl`.
- "HOST IDENTIFICATION HAS CHANGED": verify first, then `ssh-keygen -R ares.cyfronet.pl`.

2. Generate keys (use Ed25519, with a passphrase)

```
ssh-keygen -t ed25519 -a 100 -f
~/.ssh/id_ed25519_ares -C "you@laptop-ares"
ssh-keygen -t ed25519 -a 100 -f
~/.ssh/id_ed25519_athena -C "you@laptop-athena"
chmod 700 ~/.ssh && chmod 600 ~/.ssh/id_ed25519_*
```

RSA (3072+ bits) only as a fallback. Never share the private key.

2b. Copy the public key to Ares, then log in with it

```
ssh-copy-id -i ~/.ssh/id_ed25519_ares.pub
plguser@ares.cyfronet.pl
# manual alternative (append, never overwrite):
scp ~/.ssh/id_ed25519_ares.pub
plguser@ares.cyfronet.pl:~/
ssh plguser@ares.cyfronet.pl 'mkdir -p ~/.ssh &&
chmod 700 ~/.ssh && cat ~/id_ed25519_ares.pub >>
~/.ssh/authorized_keys && chmod 600
~/.ssh/authorized_keys && rm ~/id_ed25519_ares.pub'
ssh -i ~/.ssh/id_ed25519_ares -o IdentitiesOnly=yes
plguser@ares.cyfronet.pl
```

Windows PowerShell has no `ssh-copy-id`: use the manual method. You can upload it to the PLGrid portal, see section 5.

3. Config (`~/.ssh/config`, first match wins, `Host *` last)

```
Host ares
    HostName ares.cyfronet.pl
    User plguser
    IdentityFile ~/.ssh/id_ed25519_ares
    IdentitiesOnly yes
Host athena
    HostName athena.cyfronet.pl
    User plguser
    IdentityFile ~/.ssh/id_ed25519_athena
    IdentitiesOnly yes
Host *
    ServerAliveInterval 60
    ServerAliveCountMax 3
    AddKeysToAgent yes
```

macOS: also add `UseKeychain yes` under `Host *` and `IgnoreUnknown UseKeychain` at the top.

4. Agent and persistence

OS	Setup
macOS	Agent runs already. `UseKeychain yes` or `ssh-add --apple-use-keychain KEY`
Linux	`eval "$(ssh-agent -s)"`; persist with the desktop keyring, systemd user unit or `keychain`
Windows PowerShell	Admin: `Set-Service ssh-agent -StartupType Automatic; Start-Service ssh-agent`, then `ssh-add $env:USERPROFILE\.ssh\id_ed25519_ares`
WSL	Own agent: use `keychain`, keep keys in WSL `~/.ssh`

Check: `ssh-add -l`

PuTTY is not covered and we discourage using it: use PowerShell (OpenSSH) or WSL.

5. Register the public key

Copy `cat ~/.ssh/id_ed25519_ares.pub` (and the Athena one) into your account at **portal.plgrid.pl**. Wait a few minutes, then `ssh ares` and `ssh athena`.

6. Best practices

Passphrase on every key, one key per service, rotate and remove old keys, `ForwardAgent` only for trusted hosts, keep screen lock and disk encryption on, verify host fingerprints.

7. Troubleshooting

Command	Use
`ssh -v ares`	see keys offered and why auth fails
`ssh -G ares`	show the effective config
`ssh -F /dev/null plguser@ares.cyfronet.pl`	bypass your config
`ssh-add -l`	is the key in the agent?

Error	Fix
`Permission denied (publickey)`	key not registered or propagated, wrong `User` or key
`Too many authentication failures`	add `IdentitiesOnly yes`
`Bad configuration option: usekeychain`	add `IgnoreUnknown UseKeychain`
`Bad owner or permissions`	`chmod 600` on config and keys
Idle disconnects	keep-alive settings